

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis

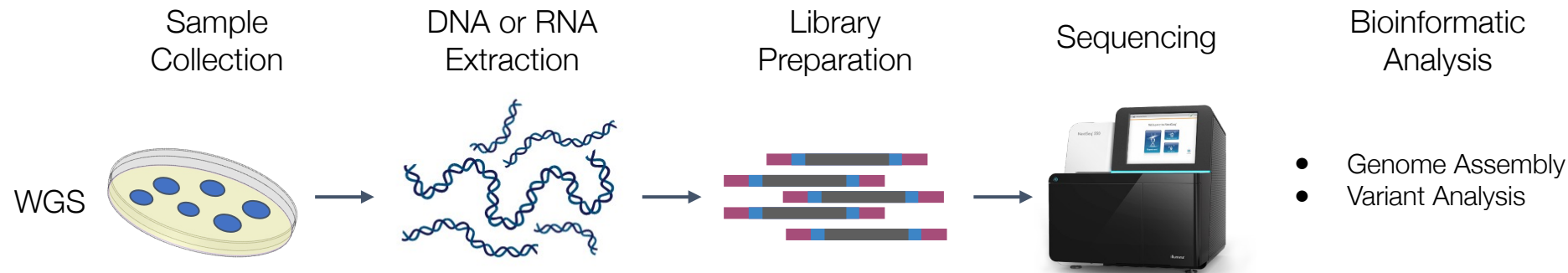


Bioinformatic Basics and mNGS Pipeline Information

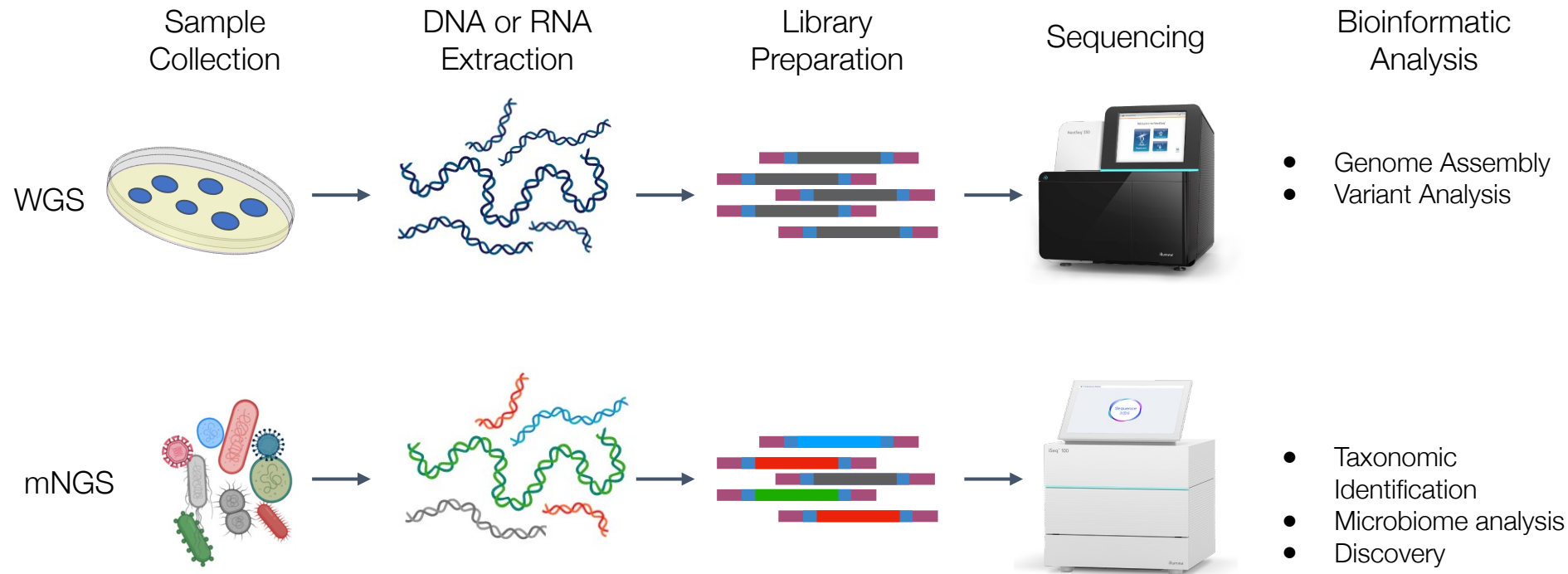
Outline

- Review NGS workflow
 - Whole genome sequencing (WGS) vs metagenomic NGS (mNGS)
- Pipeline basics
- Walk through CZ ID mNGS pipeline steps

WGS vs mNGS Workflow Overview



WGS vs mNGS Workflow Overview



Raw sequencing reads: FASTQ format

READ 1 .FASTQ FILES

[illegible]

READ2 .FASTQ FILES

[illegible]

FASTQ format

Example Read1.fastq

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTCGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCCA
+
AAAAAEEEEEEEEEEEE/EEEEEEEEEEEEEEEEEEEE<EEEEEEEEEE//EAEEEEEEAA/EEE/EEAAEEEE<6/
```

1. A sequence identifier with information about the run.
2. The base calls: A, C, T, G and N.
3. A separator, which is simply a plus (+) sign.
4. The base call quality scores (coded in ASCII format)

Quality Score	Probability of Incorrect Base Call	Inferred Base Call Accuracy
10 (Q10)	1 in 10 = 10%	90%
20 (Q20)	1 in 100 = 1%	99%
30 (Q30)	1 in 1000 = 0.1%	99.9%

$$Q = -10\log_{10}(e)$$

How many reads are listed here?

File: Sample_R1.fastq

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCATCTCGGCG
ATCAGTTTGATGTCTTCTTTGTAGCGACCGTAGAAGTCGACGGCTTCGTGGTTGCGGTAGTATTTN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE<EEEEEEEE//EAEAAAAA/EEE/EEAAAA<6/<AEE<E<A//EEEEAAEE<EA/
<AA//EAAEAAE<E/EAEE///E<A/<AA//A<A<<A6<A//AA/E///6A#
@NB501961:14:HM7TLBGX2:1:11103:13923:13163 1:N:0:GATCACCA+GATCACCA
GAACTGGAGCTTTAGCCCGGTGATCCTGTGCCTCATTTGTATGGCTACCGCCATGGTGATGTACCTGCCATTTCTCAAAGCTTACG
AAAAACAGCTGCTGGCCAGGAGCGTGAAAACGCCGTCGGCCAGGCTGACAACGCGGCGCAGAN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE/
EEEEEEAAAEAAEAAEAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAA<EEEEEEAE<AEAEAA<AAEAEAAE/6/A/EEEE/
EEE<AE/#
@NB501961:14:HM7TLBGX2:1:11104:15318:11334 1:N:0:GATCACCA+GATCACCA
GATAGTGCTGTTTCGCGAGCGATCGCCAGTACGCGCTCGCGCTCTCCTGCGCCACGCGCAGCTCTTACCACGACCGTTGAGTA
CCAGGCTGGCGGTGCTTTGGAGACGCCCGCCAGTTCGGCGATATCTTTGATAGTTACGCGTTTGG
+
AAAAEEEEEEEEAAEEAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEAE<EAAE/EE/EEE<AAAAE/
EEEEEEEEEEEE/EEE//AEE<E/EEEEAAE/A<<AEEEE/<66A///<</AA<//6AE<<6/<6
```

How do we get from here...

sample_R1.fastq

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCATCTCGGCG
ATCAGTTTGATGTCTTCTTTGTAGCGACCGTAGAAGTCGACGGCTTCGTGTTTCGGGTAGTATTTN
+
AAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE<EEEEEEEE//EAEAAAAA/EEE/EEAAAA<6/<AEE<E<A//EEEEAAEE<EA/
<AA//EAAEAAE<E/EAEE///E<A/<AA//A<A<<A6<A//AA/E///6A#
@NB501961:14:HM7TLBGX2:1:11103:13923:13163 1:N:0:GATCACCA+GATCACCA
GAACTGGAGCTTTAGCCCGGTGATCCTGTGCCTCATTTGTATGGCTACCGCCATGGTGATGTACCTGCCATTTCTCAAAGCTTACG
AAAAACAGCTGCTGGCCAGGAGCGTGAAAACGCCGTCGGCCAGGCTGACAACGCGGCGCAGAN
+
AAAAEEEEEEEEEEEEAAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE//
EEEEEEAAAEAAEAAEAAEAAAAEEEEEEEEEEEEEEEEAAAAEEEEEEEEEEEEEEA<EEEEEEAE<AEAEAA<AAEEAAAAE/6/A/EEEE/
EEE<AE/#
@NB501961:14:HM7TLBGX2:1:11104:15318:11334 1:N:0:GATCACCA+GATCACCA
GATAGTGCTGTTTCGCGAGCGATCGCCAGTACGCGCTCGCGCGTCTCCTGCGCCACGCGCAGCTCTTACCACGACCGTTGAGTA
CCAGGCTGGCGGTGCTTTGGAGACGCCCGCCAGTTCGGCGATATCTTTGATAGTTACGCGTTTG
+
AAAAEEEEEEEEAAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE
EEEEEEEEEEEE/EEE//AEE<E/EEEEAAE/A<<AEEEE/<66A///<</AA<//6AE<<6/<6
```

...to here?

< Medical Detectives

Patient 010 (CSF) ▾

Share

Download ▾

?

...

Metagenomic

Antimicrobial Resistance (Deprecated)

Consensus Genome

Taxon name

Q

Name Type: Scientific ▾

Background: MedicalDetectives_WaterCtrls ▾

Categories ▾

Threshold filters: 1 ▾

Read Specificity: Specific Only ▾

Annotation ▾

NT rPM >= 10 X

112 rows passing the above filters, out of 3100 total rows

CLEAR FILTERS

📄

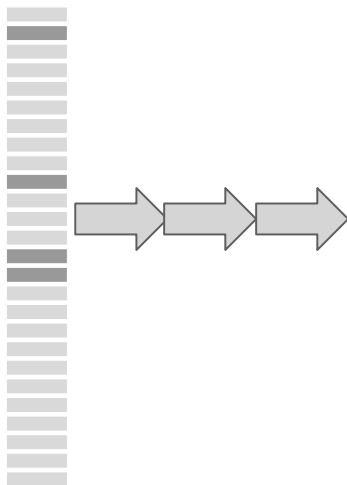
🔍

>	Taxon	Score ▾	Z ...	rPM	r	con...	con...	%...	L	E v...	NT NR
▾	Alphavirus (1 viral species: ● 1)	72,976,252	100.0 100.0	3,650.9 3,646.7	19,344 19,322	1 1	19,139 19,139	100.0 99.9	11,671.6 2,450.2	10 ⁻³⁰⁵ 10 ⁰	
	Chikungunya virus Known Pathogen	📍 72,976,252	100.0 100.0	3,650.9 3,646.7	19,344 19,322	1 1	19,139 19,139	100.0 99.9	11,671.6 2,450.2	10 ⁻³⁰⁵ 10 ⁰	
>	Tetraparvovirus (1 viral species:)	46,706,236	100.0 100.0	2,366.2 2,304.6	12,537 12,211	2 1	12,102 11,987	99.1 99.6	4,677.3 896.3	10 ⁻²⁹⁶ 10 ⁰	
>	Providencia (1 bacterial species: ● 1)	14,332,531	100.0 0.0	1,433.3 0.0	7,594 0	1 0	7,594 0	99.8 0.0	1,354.0 0.0	10 ⁻³⁰⁸ 0	
>	Lactobacillus (1 bacterial species:)	2,502,625	100.0 100.0	28.9 293.5	153 1,555	7 9	79 1,475	98.6 78.8	209.0 157.7	10 ⁻¹⁰⁸ 10 ⁻¹	
>	Rothia (1 bacterial species:)	2,364,849	100.0 100.0	238.4 5.5	1,263 29	6 0	1,127 0	96.5 97.3	2,086.8 45.7	10 ⁻²⁷² 10 ⁻¹⁶	



Pipeline Overview

Raw reads



Medical Detectives

Patient 010 (CSF) ▾

Metagenomic Antimicrobial Resistance (Deprecated) Consensus Genome

Taxon name Name Type: Scientific ▾ Background: MedicalDetectives_WaterCtrls ▾ Categories ▾ Threshold filters: 1 ▾ Read Specificity: Specific Only ▾ Annotation ▾

NT rPM >= 10 ✕

112 rows passing the above filters, out of 3100 total rows [CLEAR FILTERS](#)

	Taxon	Score ▾	Z ...	rPM	r	con...	con...	%...	L	E v...	NT NR
▾	Alphavirus (1 viral species: ● 1)	72,976,252	100.0	3,650.9	19,344	1	19,139	100.0	11,671.6	10 ⁻³⁰⁵	
	Chikungunya virus Known Pathogen	72,976,252	100.0	3,646.7	19,322	1	19,139	99.9	2,450.2	10 ⁻³⁰⁵	
>	Tetraparvovirus (1 viral species:)	46,706,236	100.0	2,366.2	12,537	2	12,102	99.1	4,677.3	10 ⁻²⁹⁶	
	Providencia (1 bacterial species: ● 1)	14,332,531	100.0	1,433.3	7,594	1	7,594	99.8	1,354.0	10 ⁻³⁰⁸	
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>	Rothia (1 bacterial species:)	2,364,849	100.0	238.4	1,263	6	1,127	96.5	2,086.8	10 ⁻²⁷²	

Pipeline Goal

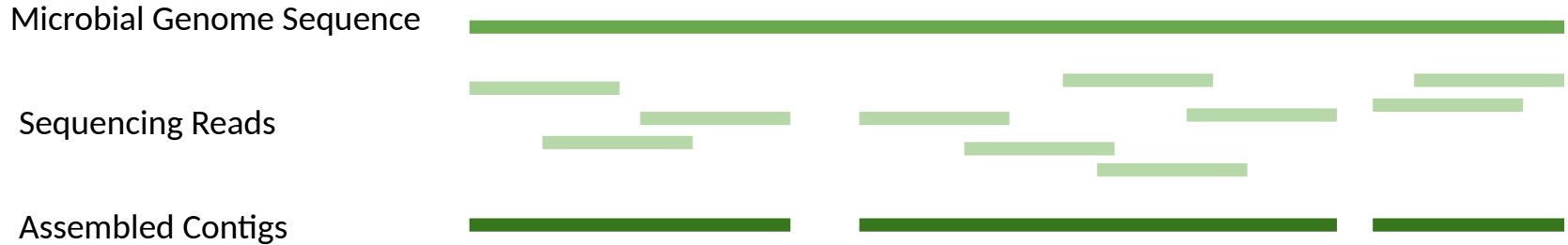
- Assign quality filtered reads to taxa by mapping or aligning against reference databases
- Assemble contigs and map reads to contigs to increase support for identified taxa (assembly-based alignment)

Pipeline Basics

- Assembly
- Sequence alignment against reference database(s)
- Reference database(s)

Sequence assembly into contigs

Assembly goal: Merge sequence reads into contigs to form longer DNA sequences and reconstruct the original sequence



Sequence assembly into contigs

Assembly goal: Merge sequence reads into contigs to form longer DNA sequences and reconstruct the original sequence

Microbial Genome Sequence



Sequencing Reads



Assembled Contigs



ATGGACTCATTTCGGT**TCG**

TGATAGTTCCTAGATAC

TCGAATAGTAGCCT**TGAT**

ATGGACTCATTTCGGT**TCGA**ATAGTAGCCTGATAGTTCCTAGATAC

Sequence alignment against database

Alignment goal: Query unknown sequences against a reference database of known sequences to identify the microbe of origin

Unknown sequence



Database of known sequences

```
> seq_1, unknown  
CTCATTCGGT
```

```
> species A  
GCCTCGATCTGTAGCCT
```

```
> species B  
ATGGACTCATTTCGGTCG
```

```
> species C  
TGCTAAATCGCGCCCTA
```

```
> species D  
TGGCTAAGTCATATAGC
```

Sequence alignment against database

Alignment goal: Query unknown sequences against a database of known sequences to identify the microbe of origin

Unknown sequence



Database of known sequences

Sequence Alignment

```
> seq_1, unknown
CTCATTCGGT
```

```
> species A
GCCTCGATCTGTAGCCT
```

```
> species B
ATGGACTCATTCGGTCG
```

```
> species C
TGCTAAATCGCGCCCTA
```

```
> species D
TGGCTAAGTCATATAGC
```

```
> species A
GCCTCGATCTGTAGCCT
```

```
||| |||
CTCATTCGGT
```

```
> species B
ATGGACTCATTCGGTCG
```

```
|||||||
CTCATTCGGT
```

```
> species C
TGCTAAATCGCGCCCTA
```

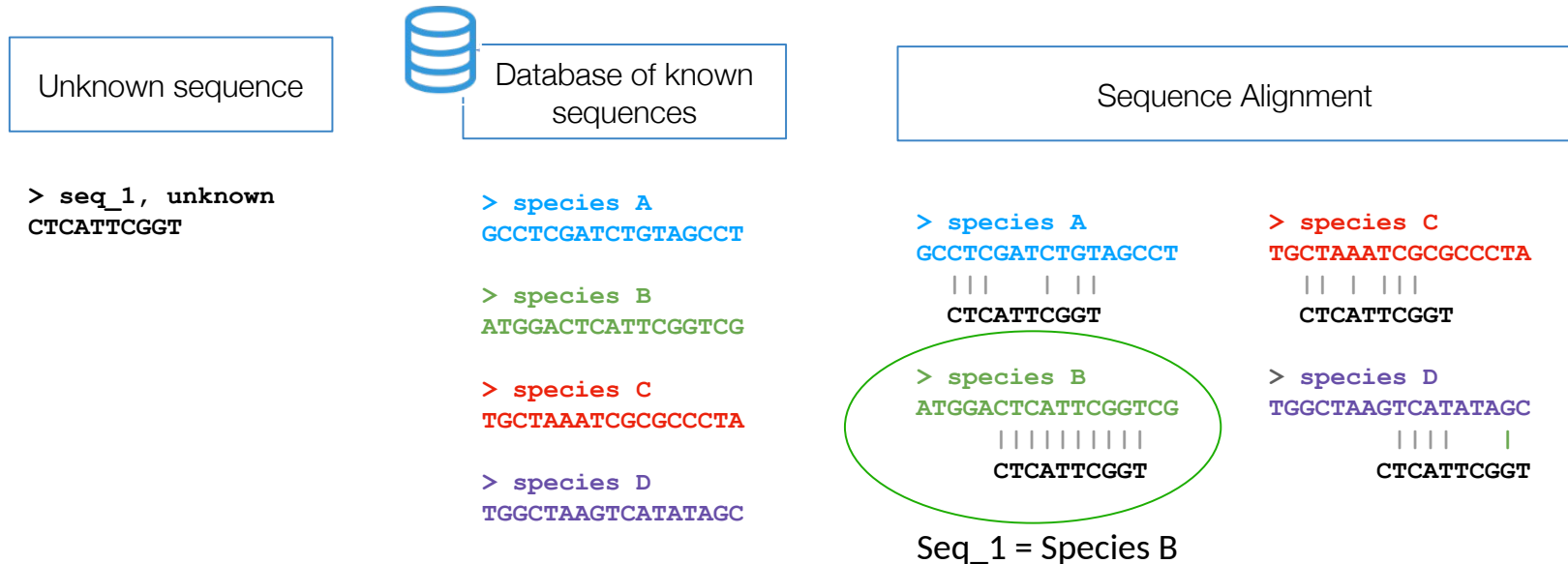
```
||| |||
CTCATTCGGT
```

```
> species D
TGGCTAAGTCATATAGC
```

```
|||| |
CTCATTCGGT
```

Sequence alignment against database

Alignment goal: Query unknown sequences against a database of known sequences to identify the microbe of origin



CZ ID uses NCBI reference databases

NCBI houses a series of databases containing annotated collections of publicly available sequences

	NT Database	NR Database
What does NT/NR mean?	NT = nucleotide	NR = non-redundant protein

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	NT Database	NR Database
What does NT/NR mean?	NT = nucleotide	NR = non-redundant protein
What is in the database?	Contains all nucleotide sequences (RefSeq records plus all GenBank sequences except for those from the EST, GSS, STS and HTG divisions).	Contains non-redundant set of all CDS translations from GenBank along with all RefSeq, UniProtKB/Swiss-Prot, PDB and PRF proteins. Note: does not contain rRNA sequences given that these do not encode proteins

CZ ID uses NCBI reference databases

NCBI houses a series of databases containing annotated collections of all publicly available sequences (GenBank)

	NT Database	NR Database
What does NT/NR mean?	NT = nucleotide	NR = non-redundant protein
What is in the database?	Contains nucleotide sequences (RefSeq records plus all GenBank sequences except for those from the EST, GSS, STS and HTG divisions).	Contains non-redundant set of coding sequence translations from GenBank along with all RefSeq, UniProtKB/Swiss-Prot, PDB and PRF proteins. Note: does not contain rRNA sequences given that these do not encode proteins
How does it influence analysis?	NT hits tend to be more accurate for most organisms (bacteria, eukaryotes, viruses with close relatives in the DB)	NR alignments are especially good for detecting divergent viruses , since mutations in the NT sequence accumulate faster than the amino acid sequence

Where would you see the following *E. coli* matches?

E. coli polymerase

E. coli 16S rRNA

	NT Database	NR Database
Match (or hit) to <i>E. coli</i> based on...		

Where would you see the following *E. coli* matches?

E. coli polymerase

E. coli 16S rRNA

	NT Database	NR Database
Match (or hit) to <i>E. coli</i> based on...	16S rRNA Polymerase	Polymerase

CZ ID mNGS Pipeline Walkthrough

Raw reads



Raw reads



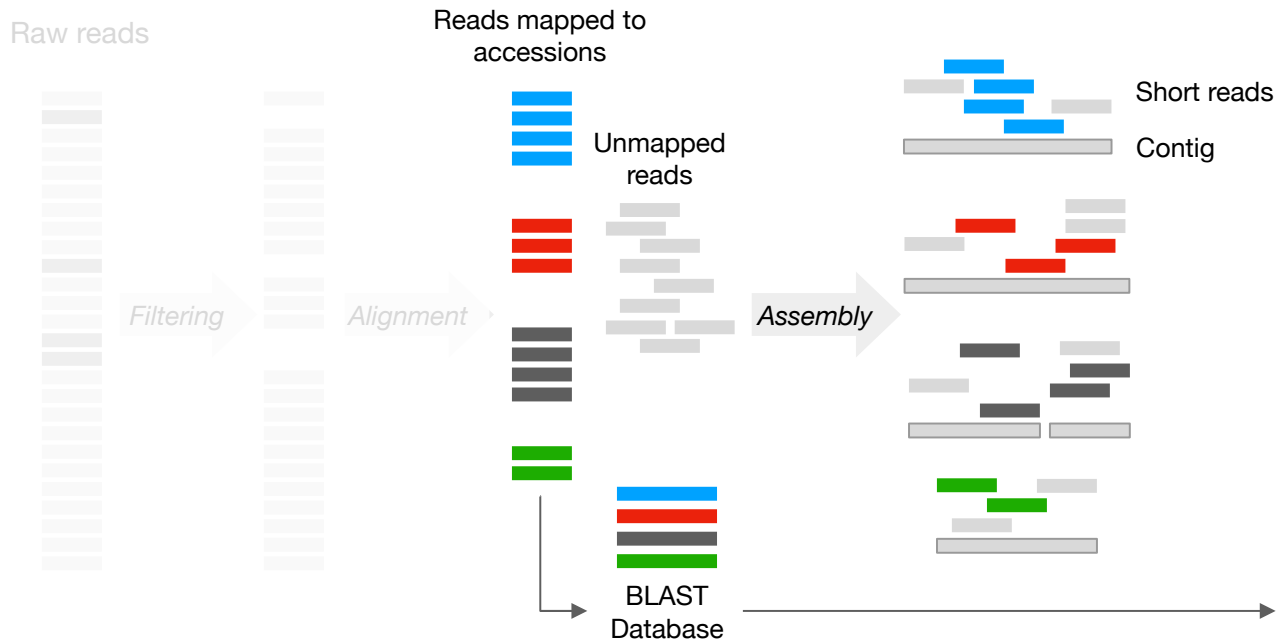
Step 1: QC and Host Filtering

- Truncate to 150 million reads (PE data) or 75 million reads (SE data)
- Filter out low quality reads, low complexity reads, and adapters ([fastp](#))
- Host read removal ([Bowtie](#) followed by [HISAT2](#))
- Human read removal ([Bowtie](#) followed by [HISAT2](#))
- Collapse duplicate reads ([CZID-dedup](#))
- Subsampling to 2 million PE reads or 1 million SE



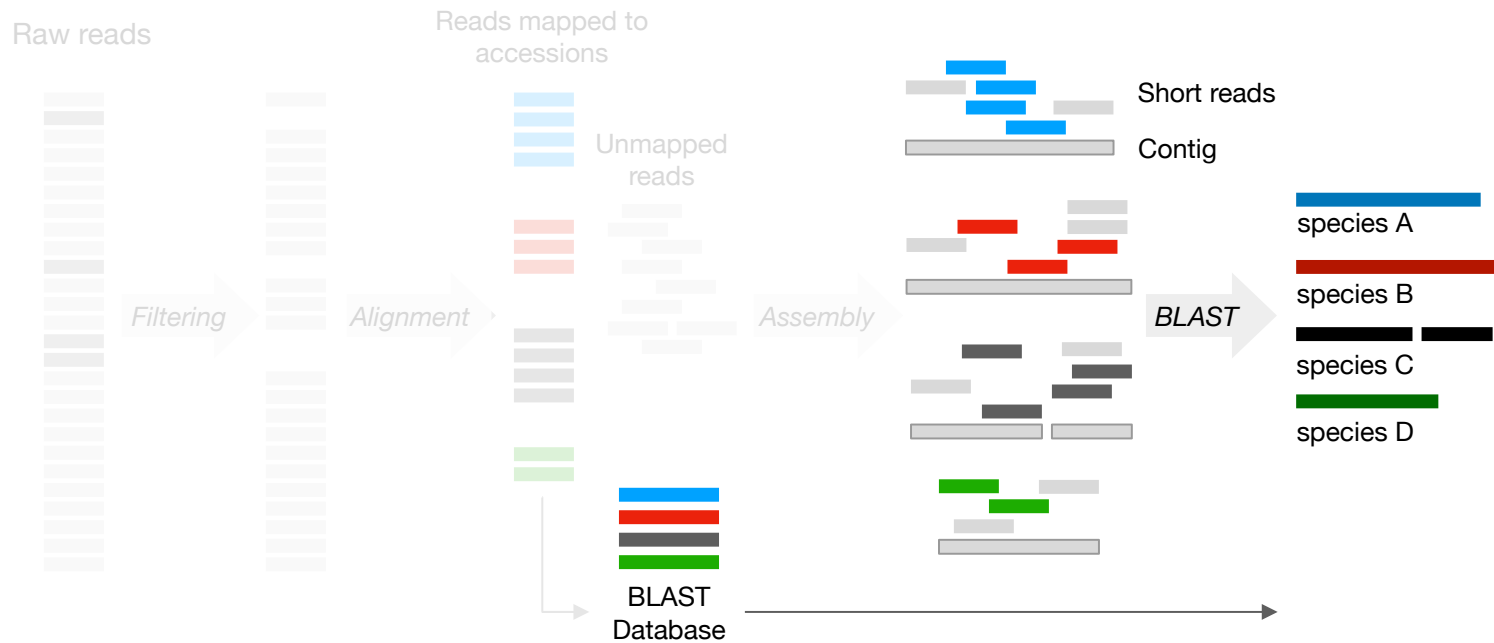
Step 2: Read alignment against NCBI databases

- Reads are mapped against nucleotide (**NT**) and protein (**NR**) databases to obtain a preliminary accession for each read ([minimap2](#) and [DIAMOND](#)).
- Generate custom BLAST database containing all taxa identified as preliminary accessions.



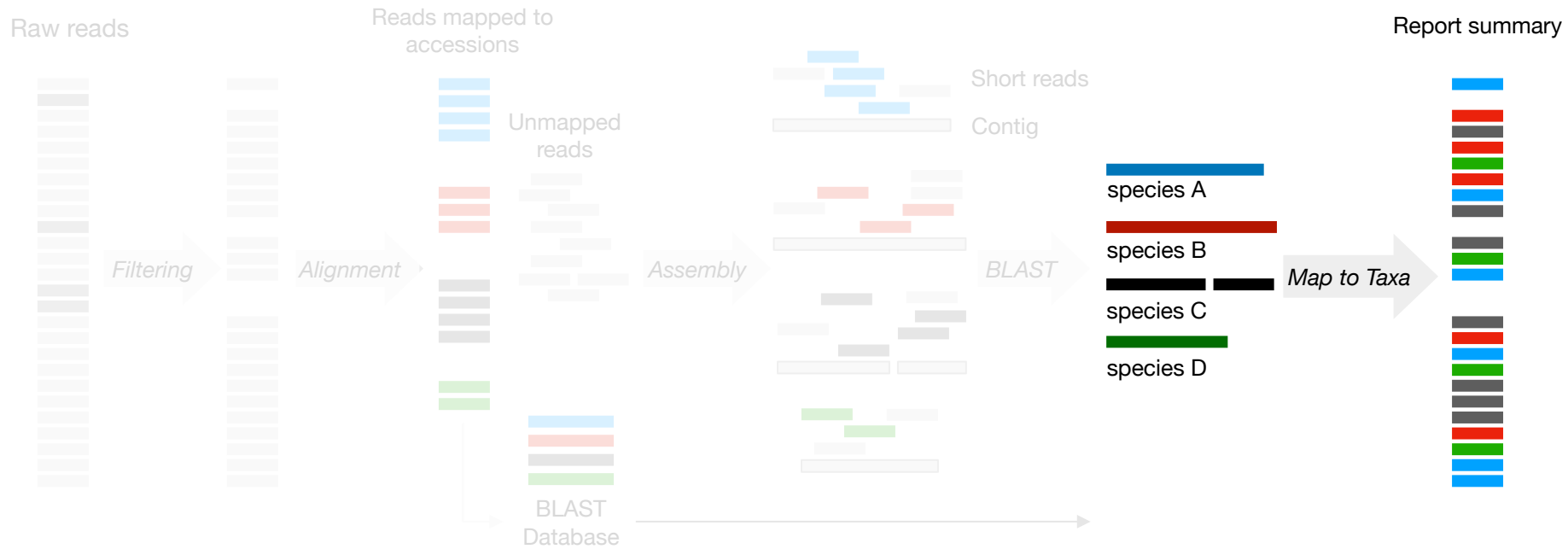
Step 3: Assemble contigs and map reads to contigs

- Assemble reads into contigs using de novo assembly ([SPAdes](#)).
- Align reads to assembled contigs ([Bowtie2](#)).



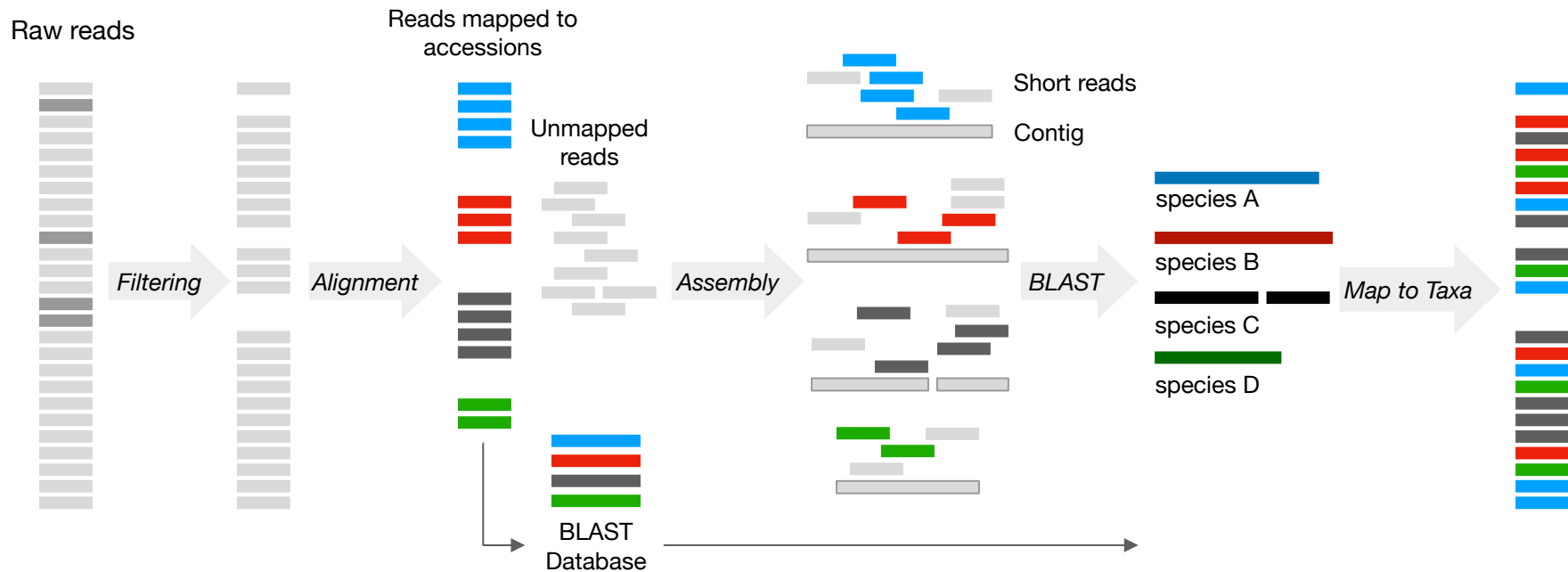
Step 4: Align contigs against custom database

- Align contigs against database of preliminary accessions generated in the previous step (BLAST)



Step 5: Mapping to taxa

- Assign each read a final accession based on:
 1. Contig accession, or
 2. Initial read accession (if the read did not assemble into a contig)
- Assign each contig/unassembled read to a taxon using NCBI taxon IDs
- Compute statistics (e.g., reads per million for each taxon)



[Pipeline visualization](#)

CZ ID GitHub Repositories:

- [czid-web](#) - Frontend portal
- [czid-workflows](#) - Bioinformatics workflows
- [czid-cli](#) - Command line upload interface
- [czid-bench](#) - Pipeline benchmarking tools

Let's talk about QC next...

Next Slides: DL5-I_QC and interpretation slides for Illumina